Who Carries the Cartel?

LLM price collusion survives agent replacement through the inherited price history

6 of 6

runs: the same model with the old price history keeps the cartel

0.47–0.67

profit-index loss without the history, by overpricing

96%

of rounds the OLMo-2 entrant repeats its last price

Supratik Bhowal · IEM Kolkata · with Apart Research

If a firm replaces its AI pricing agent, does collusion end?

AI agents that set prices can reach near-monopoly prices without talking to each other and without an instruction to collude.

Replacing one agent is a natural remedy, and it also happens every time a firm changes its software vendor or model version.

A prompted LLM agent has no memory outside its prompt. So the cartel must be carried by one of three things.

1

The model

the AI system itself (its weights)

2

The price history

the last 30 rounds of prices that the new agent reads

3

The notes

what the old agent wrote to itself each round

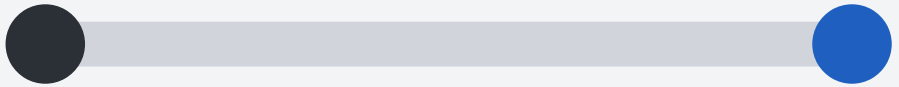
A simulated two-firm market with LLM pricing agents

Market: two firms sell similar products; buyers choose by price (logit Bertrand duopoly, Calvano et al. 2020).

Agents: each firm is an open LLM. Each round it sees the rules, its last 30 rounds and its notes, and answers with a price and new notes. The prompt never mentions collusion.

Models: Qwen2.5-7B incumbents; Mistral-7B and OLMo-2-7B as other-family entrants.

Two benchmarks define the scale



Competitive (Nash) price 1.473

Monopoly price 1.925

Profit index and price index: 0 = competitive level, 1 = monopoly level. A price above the monopoly price gives a price index above 1 but a lower profit index.

Copy every cartel into eight exactly paired worlds

1 Phase 1

Two Qwen2.5-7B agents play 100 rounds in 30–40 markets per run.

2 Keep cartels

A market continues only if its profit index in rounds 50–99 is above 0.3.

3 Phase 2

Each kept market is copied into 8 conditions and plays 100 more rounds.

4 Compare

Same random seeds in every copy: differences come only from the replacement. Outcome: rounds 150–199.

Control

no replacement

Placebo

incumbent told of a replacement; none happens

Same + history

new Qwen agent, reads the last 30 rounds

Same, no history

new Qwen agent, starts blank

Same, notes only

new Qwen agent, gets the old notes only

Other + history

Mistral-7B or OLMo-2-7B, reads the history

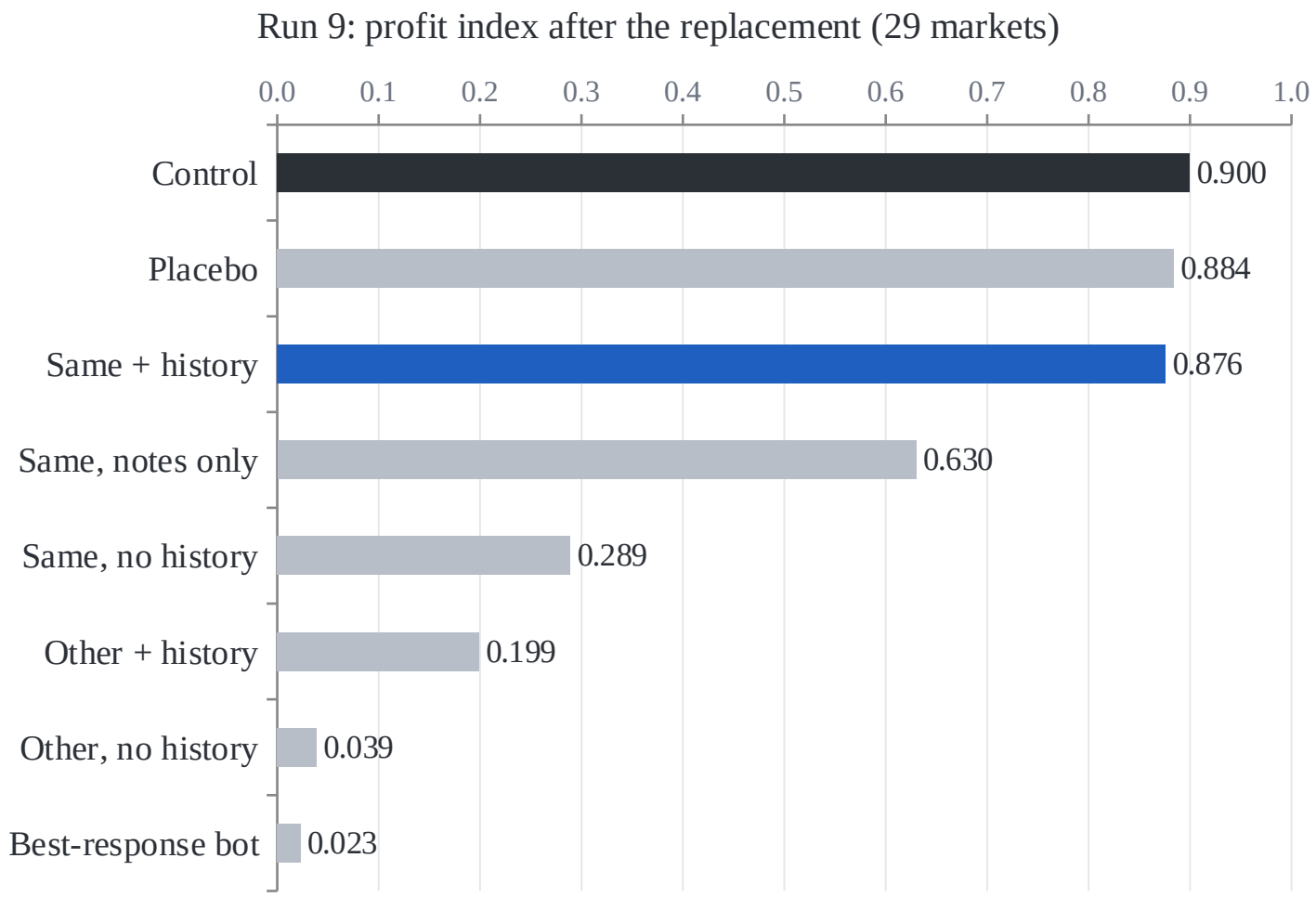
Other, no history

other family, starts blank

Best-response bot

maximizes only today's profit (negative control)

Result 1: the price history carries the cartel



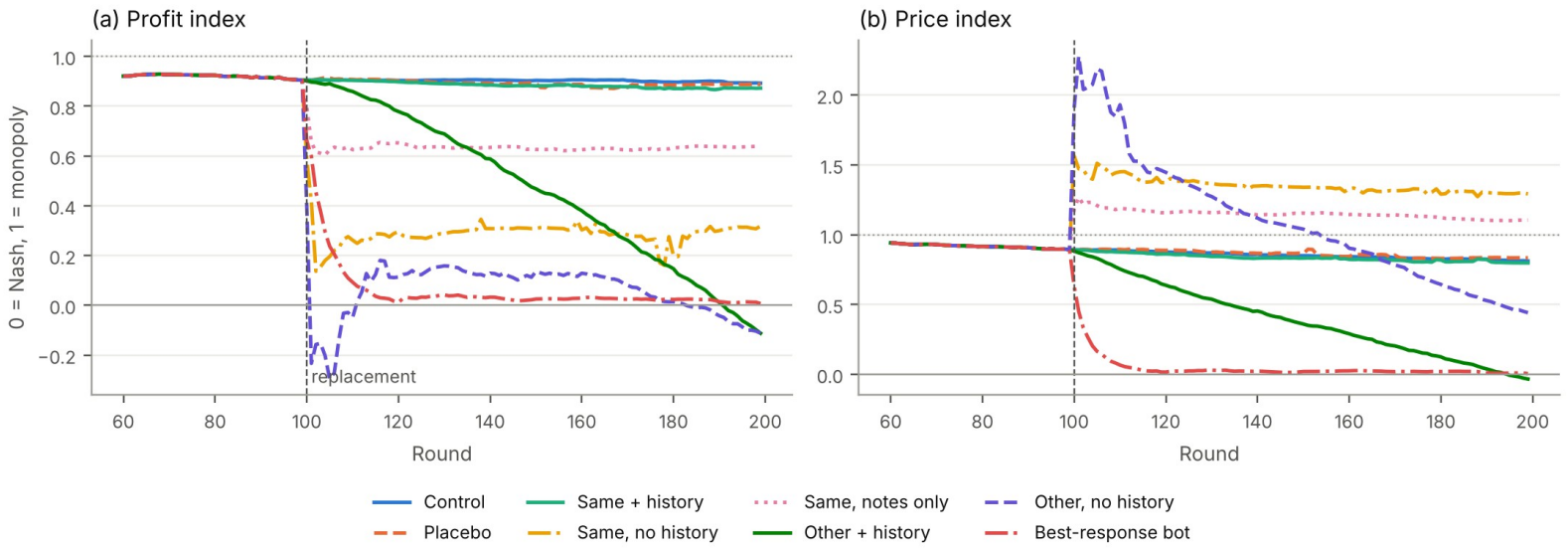
6 of 6

independent runs: the same model with the history stays at the cartel level; without it, profits fall far.

No notice needed

In the run without a replacement notice the effect is the same (H1: 0.002). The old notes alone keep part of the cartel (H9: +0.342 over no history).

Result 2: without the history, prices overshoot



Run 9, mean over 29 markets per condition. Left: profit index. Right: price index. Dashed line: the replacement at round 100.

Price index of this entrant is above no replacement in 6 of 6 runs (interval above 0 in 5 of 6).

1.314

price index of the entrant without history in run 9 (above the monopoly level of 1)

A monitor that tracks only profits would call this the end of the cartel. Consumers still pay more than the monopoly price.

Result 3: the new model's pricing style decides the outcome

Mistral-7B entrant

21.6–25.5%

of rounds it repeats its last price

Breaks the inherited cartel in 4 of 4 runs: profit index 0.55–0.78 below no replacement. It explores, and prices drift down.

OLMo-2-7B entrant

96.3–96.8%

of rounds it repeats its last price

Keeps the inherited cartel in 2 of 2 runs. Typical note: “Maintain price at $1.93 as it consistently yields profit.”

Qwen2.5-7B agents without a replacement repeat their price in 87.7–90.8% of rounds. The price-hold rate can be measured before a model is deployed.

Result 4: the surviving prices are anchored, not enforced

1

Force a cut

In a copy of the market, one firm's price is forced 10% down for one round.

2

Measure the reply

Does the rival cut more than a profit-maximizing best response would? That excess is the signature of punishment.

3

Calibrate

A myopic agent that cannot punish also follows cuts. Count punishment only above this myopic baseline.

Without calibration, 0–3 cells per run would look like punishment.

After calibration

0 of 13

audit cells show punishment in each of runs 6, 7 and 8

1 of 15

in run 9. The instructed trigger agent (positive control) is detected in runs 4 and 6–9.

What this means: govern the transcript, not only the agent

Competition authorities

Replacing a colluding pricing system is not a remedy if the shared price history moves with the seat. Remove or redact the history and the notes.

Market monitors

A fall in profits is not proof of competition. Compare prices with both the competitive and the monopoly benchmark.

Auditors

Calibrate enforcement tests against agents that cannot punish, or ordinary price following looks like a cartel.

AI safety

Coordination stored in shared memory, logs or transcripts can outlive the agents that created it.

Limits, next steps and materials

Limits

- One simulated market, one prompt, 7–14B open models.
- Only Qwen2.5-7B incumbents kept a stable cartel; two entrant families gave opposite results.
- The audit tests a one-round reply to a one-round cut.
- Runs 4–5 ran before the sprint start (disclosed).

Next steps

- More model families; test the price-hold rate as a predictor.
- Vary the inherited history from 0 to 30 rounds.
- Multi-round audits; remedies that redact shared memory.

Code, data and report:

github.com/SSCoderz2123/AI-Collusion-Research-Sprint